

NONCOOPERATIVE GAME THEORY

RATIONALITY IN INTERACTION

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Noncooperative Game Theory : Rationality in Interaction

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Preface

Acknowledgement

This book is written using $\text{\LaTeX}$. The template is based on Agni Datta's "[Preprint Book Manuscript Template](#)", which I have modified to fit my requirements. The famous [ElegantLaTeX](#) template has also inspired me greatly. The source code is available in [my GitHub repository](#) for reference.

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Chapter 1

Introduction

In decision theory, we're modeled as acting as if we were maximizing our utility given some exogenous parameters. An example is that producers in competitive markets choose the output that maximizes their profit. In this setting, no individual producer's behavior affects market structure, and expectations about other producers do not influence decision-making.

However, such perspectives of analysis is overly restriction and the real world is not always the case. For example, collusion requires participants to act based on the expectation of other firms' behavior. This is what game theory studies. Game theory is a subject that deals with interaction among decision-makers, focusing on the existence and properties of equilibrium. Equilibrium is a kind of stability, where no participants have incentive to deviate. We'll formalize this concept later on.

1.1 Game

1.1.1 Basic Composition of a Game

Now we start to consider the formal definition of games. We try to formalize these concepts by analyzing the game of rock-scissors-pape.

Example 1.1.1 (Rock-scissors-pape). In this game, we have 2 players and the

consequences can be represented in **Table 1.1**, where 1 stands for win, 0 stands for draw and -1 stands for lose. This is also called the payoff matrix, which is widely used in game theory. The embedding assumption of this game is that players prefer winning to losing, inducing a preference relation.

		Player 2		
		R	S	P
Player 1	R	0, 0	1, -1	-1, 1
	S	-1, 1	0, 0	1, -1
	P	-1, 1	1, -1	0, 0

Table 1.1: Outcome of the Rock-scissors-pape Game

A game consists of players (along with their preferences) and rules (that maps actions to consequences). Each player i , denoted by A_i , has a set of available actions, called action set $\mathcal{A}_i$. We use a_i to denote a certain action, i.e., $a_i \in \mathcal{A}_i$. As for "rule", it maps the action profile $a = (a_i)_{i \in N}$ to a consequence c , i.e., a mapping g from the Cartesian product of action sets $\mathcal{A} = \prod_{i \in N} \mathcal{A}_i$ to the consequence set C . If we assume the participants are rational, they can surely compare among 2 consequences, inducing a preference $\leq_i^*$ of player i over $C \times C$.

Definition 1.1.1 (Game). A game is a 4-tuple $\langle N, \mathcal{A}, g, \{\leq_i^*\}_{i \in N} \rangle$, where N is the set of player index. If $\leq_i^*$ admit a utility representation, we may also write $\langle N, \mathcal{A}, g, \{u_i\}_{i \in N} \rangle$ instead.

Notice that the consequence and the action is associated with a mapping, we can equivalently define the preference relation over $\mathcal{A} \times \mathcal{A}$. Let a_{-i} denote the action of other players, i.e., $a_{-i} = (a_j)_{j \in N \setminus \{i\}}$, then the preference of player i can be defined as

$$(a_i^{(1)}, a_{-i}^{(1)}) \leq_i (a_i^{(2)}, a_{-i}^{(2)}) : \iff g(a_i^{(1)}, a_{-i}^{(1)}) \leq_i^* g(a_i^{(2)}, a_{-i}^{(2)}).$$

Then a game can be represented by $\langle N, \mathcal{A}, \{\leq_i\}_{i \in N} \rangle$.

Such a game is of the simplist form since it's almost one-shot and static. We'll introduce history when it comes to the extensive form.

1.1.2 Strategic Form

A strategic game is a scenario where players make decisions independently and simultaneously.

To understand the meaning of “independently and simultaneously”, let’s consider an experimental setting, every subject, anonymously sitting behind a computer, is shown a payoff matrix and asked to choose their strategy*. They won’t be able to see the outcome until everyone makes their choice. Two important things to understand:

- They don’t necessarily to make decisions at exactly the same time.
- It’s not necessarily a one-shot game. It can be repeated, but across rounds, no strategic links are allowed.

To make a correct decision, they should care not only about their own choice, but also make decisions under other subjects’ possible action, i.e., form a belief. To be more specific, we need to consider two things: how should one interact given opponents’ action? And, what are opponents’ possible actions? We will formalize this idea soon after we give the definition of a strategic game.

Definition 1.1.2 (Strategic Game). A strategic game is a 3-tuple $\langle N, \mathcal{A}, \{\leq_i\}_{i \in N} \rangle$. Everything is defined in the introduction part of the book. If every $\mathcal{A}_i$ is finite, we say the game is finite.

* Note that this is the perfect information case. We’ll talk about the impact of information in 1.2.4.

1.1.3 Extensive Form**1.1.4 Common Example of Games****1.2 Belief, Information and Knowledge****1.2.1 State and Belief****1.2.2 From State to Information****1.2.3 Knowledge and Common Knowledge****1.2.4 Perfect and Incomplete Information****1.3 Decision Under Uncertainty**

Part I

Strategic Games

Before getting into the definitions, we make clear what our goal is in strategic games as a player. As we assume players are rational, i.e., payoff seekers, we are solving a maximum problem. Given different game settings or solution concepts, there's slight difference in notation or technique. But, generally speaking, a player in a game should form a reasonable belief and make optimal decisions under such a belief.

In general, we consider that a player i with the belief $(\mathcal{A}_{-i}, \pi_i)$, where π_i is a probability measure on $\mathcal{A}_{-i}$, is trying to maximize his/her payoff by choosing a strategy $\sigma_i \in \Delta(\mathcal{A}_i)$, i.e.,

$$\max_{\sigma_i \in \Delta(\mathcal{A}_i)} \sum_{a_i \in \mathcal{A}_i} \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma_i(a_i) \pi_i(a_{-i}) u_i(a_i, a_{-i}), \quad (1.1)$$

and denote its solution the best response correspondence

$$BR_i(\pi_i) := \arg \max_{\sigma_i \in \Delta(\mathcal{A}_i)} \sum_{a_i \in \mathcal{A}_i} \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma_i(a_i) \pi_i(a_{-i}) u_i(a_i, a_{-i}). \quad (1.2)$$

As long as the belief is reasonable, a game is trivial as player can do their best by playing a strategy in the best-response correspondence. Thus, we'll first formalize the reasoning process and then turn to solve such a UMP.

For the complete information case, we'll first introduce several settings where belief is determined in different ways, and under such belief, how should players react. In the following section, we introduce the concept of rationalizability, which concerns what belief is reasonable.

Chapter 2

With Perfect Information

2.1 Equilibrium

In this section, we make clear, given a reasonable belief, how you should behave.

2.1.1 Pure Strategy Nash Equilibrium

2.1.1.1 Definition

There're games that people may stick to one single strategies, see the following payoff matrix. No matter it's player 1 or 2, he/she should only choose action

		Player 2	
		A	B
Player 1	A	2, 2	0, 0
	B	0, 0	0, 0

Table 2.1: A Coordination Game

A as that's the only strategy that gives them a positive payoff. We refer to this simplification of strategy a pure strategy.

Definition 2.1.1 (Pure Strategy). A pure strategy is a strategy that assigns probability one to a single action.

We assume that every player is palying a pure strategy, then we can simplify (1.1) and (1.2) as

$$\max_{a_i \in \mathcal{A}_i} u_i(a_i, a_{-i})$$

$$BR_i(a_{-i}) = \arg \max_{a_i \in \mathcal{A}_i} u_i(a_i, a_{-i}).$$

We refer to the steady state where every subject is playing a best-response equilibrium.

Definition 2.1.2 (Pure Strategy Nash Equilibrium(PSNE)). Denote $BR(a) := \prod_{i \in I} BR_i(a_{-i})$. We say a^* is a PSNE if $a^* \in BR(a^*)$. Or equivalently, a^* is a PSNE if

$$\forall i \in N : (a_i, a_{-i}^*) \leq (a_i^*, a_{-i}^*), \forall a_i \in \mathcal{A}_i.$$

This leads to an important way of finding the equilibrium:

- STEP I: find the best-response correspondence of each player.
- STEP II: find the action profile satisfying $a_i^* \in BR_i(a_{-i}^*)$.

2.1.1.2 Examples

BoS; Coordination Game; PD; H-D; Pennis(strictly competitive game)

2.1.1.3 Existence

As we've already seen, not every strategic game has a PSNE. In this subsubsection, we provide some conditions that guarentee the existence. By the definition of PSNE, it's natural to use the correspondence version of fixed point theorem. Here we use Kakutani's fixed point theorem, which says that as long as the correspondence is nonempty, closed and convex, there exists a fixed point.

To suffice this condition, we assume the followings:

- Action sets are nonempty compact convex subset of $\mathbb{R}^n$;
- The preference relation is continuous and quasi-concave on $\mathcal{A}_i$.

Then by Kakutani's fixed point theorem, there exists PSNE.

Theorem 2.1.1. *For a strategic game $\langle N, \mathcal{A}, \{\leq_i\}_{i \in N} \rangle$, suppose for all $i \in N$, $\mathcal{A}_i$ is nonempty, compact and convex. Further, we suppose $\leq_i$ is continuous and quasi-concave on $\mathcal{A}_i$, then there exists PSNE.*

2.1.2 Mixed Strategy Equilibrium

For many games, sticking with one strategy is somehow stupid. It's like keep playing scissor in the rock-scissor-pape game, which is clearly unwise. So, players may deliberately randomize the strategy played.

Other motivating example concerning the introduction of mixed strategy includes the matching pennis, which is similar to the rock-scissor-pape game, and the relationship between auditors and taxpayers, see [Osborne and Rubinstein \(1994, p. 37\)](#).

The idea above is to give the strategies a probability measure, which introduce the randomness of the strategies being played.

Definition 2.1.3 (Mixed Strategy). Let $\Delta \mathcal{A}_i$ denote the set of probability distribution on $\mathcal{A}_i$. Then $\alpha_i \in \Delta \mathcal{A}_i$ is a mixed strategy.

Next, we formalize the payoff. Here we use von Neumann-Morgenstern's theory on uncertainty

2.2 Rationalizability

The idea of rationalizability is that, take a player's view, not only I'm rational, my opponents are also rational. That is to say, any strategy that is not in the best-response correspondence won't be taken. Thus, I should assign probability zero to such strategies in my belief. Also, this is at least mutual knowledge, i.e., everyone in the game knows that every other player's strategy is a best-response.

Chapter 3

With Incomplete Information

Part II

Dynamic Games with Perfect Information

Part III

Dynamic Games with Incomplete Information

Appendix A

Notation

Appendix B

Supplementary Scripts

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